AI / MACHINE LEARNING INTERNSHIP PROJECT

Big Mart Sales Prediction

A supervised machine-learning regression system for estimating item outlet sales.

Submitted by: _____

Institute / Department: _____

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CERTIFICATE

This is to certify that the project titled “Big Mart Sales Prediction” has been prepared as part of the AI internship project work. The implementation, documentation and results described in this report are based on the supplied project data and the experiments performed for this submission package.

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Date: _____ Institute Seal: _____

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ABSTRACT

A supervised machine-learning regression system for estimating item outlet sales. This report explains the problem, dataset, preprocessing, model design, implementation, evaluation and limitations. The system is intended for academic demonstration and learning. It should not be treated as a production system without additional validation, monitoring and domain review.

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1. Introduction

Retail organizations use historical sales information to plan inventory, replenishment and store operations. Sales prediction is a regression task because the output is a continuous numerical value.

This project studies item-level and outlet-level attributes and estimates the sales value for a product at a particular outlet. The work uses the supplied Big Mart training and test CSV files.

The implementation is designed as a reproducible academic prototype. It demonstrates data inspection, preprocessing, supervised learning, validation and batch prediction.

2. Problem Statement and Objectives

Problem statement: Given product characteristics and outlet characteristics, estimate the expected Item_Outlet_Sales value. The supplied test file contains the same predictor fields but does not contain the target.

Objectives:

- Understand the structure and quality of the retail dataset.
- Handle missing numerical and categorical values without data leakage.
- Convert categorical variables into model-readable features.
- Train a regression model and evaluate it on held-out data.
- Generate sales estimates for the supplied test dataset.

3. Literature / Background Review

Regression estimates a continuous target from one or more input variables. Linear regression provides a simple baseline, while tree-based ensemble methods can capture nonlinear relationships and interactions.

Gradient boosting builds an additive ensemble of weak learners in sequence. Each new learner improves the fit to the residual errors of the previous ensemble. Histogram-based gradient boosting discretizes feature values into bins, which can make training efficient.

Preprocessing is critical for mixed tabular data. Numeric missing values can be imputed using a median. Categorical values can be imputed using the most frequent category and represented through one-hot encoding.

4. Dataset Description

The supplied training file contains 8,523 records and 12 columns. The test file contains 5,681 records and 11 columns. The target is Item_Outlet_Sales and is present only in the training file.

Predictors include Item_Identifier, Item_Weight, Item_Fat_Content, Item_Visibility, Item_Type, Item_MRP, Outlet_Identifier, Outlet_Establishment_Year, Outlet_Size, Outlet_Location_Type and Outlet_Type.

The training data has missing values in Item_Weight and Outlet_Size. The pipeline imputes these values rather than deleting rows. This preserves the available training observations.

The data supplied with this project is treated as the authoritative dataset. No external records are merged into it.

5. Methodology

The workflow consists of data loading, exploratory inspection, train/validation split, preprocessing, model fitting, evaluation and test-file prediction.

An 80:20 hold-out split is created with random_state=42. Numeric and categorical columns are detected from their data types. The preprocessing transformer is fitted only on the training partition, then applied to the validation partition.

The numeric branch uses median imputation. The categorical branch uses most-frequent imputation followed by one-hot encoding. handle_unknown='ignore' allows categories in the test file that were not observed during fitting.

HistGradientBoostingRegressor is configured with 180 boosting iterations, learning rate 0.08 and L2 regularization 1.0. These settings are a practical starting point, not a globally optimized hyperparameter search.

6. System Design

Input layer: CSV files containing historical training observations and new test observations.

Preprocessing layer: identifies numeric and categorical columns, fills missing values and transforms categories.

Learning layer: fits a histogram-based gradient boosting regression estimator using the processed training data.

Evaluation layer: compares validation predictions with actual validation targets using R^2, mean absolute error and root mean squared error.

Output layer: writes one predicted sales value per row in Test.csv to results/Test_Predictions.csv.

7. Implementation Details

The implementation is provided as a Python script and a Jupyter notebook. The script resolves paths relative to the repository, making it portable when the project folder is cloned.

The training pipeline is split into two phases. First, a validation experiment estimates generalization performance. Second, the preprocessing and model are refitted using all labelled training rows before producing predictions for the supplied test file.

The output CSV intentionally contains a single prediction column because the supplied test file has no target values or confirmed row identifier requirement. If a platform submission template is later provided, the output column can be adapted to that template.

The notebook includes code for data inspection, preprocessing, model training, metric calculation and test-file prediction.

8. Experimental Setup

The experiment uses Python with pandas, NumPy and scikit-learn. The random seed is fixed at 42 for the split and model to make the validation experiment reproducible.

The validation set is not used to fit preprocessing statistics or model parameters. This separation reduces optimistic evaluation caused by information leakage.

Evaluation metrics: R^2 measures explained variance relative to a mean-target baseline; MAE gives average absolute prediction error in sales units; RMSE penalizes larger errors more heavily than MAE.

9. Results and Discussion

The held-out validation experiment achieved $R^2 = 0.5861$, MAE = 733.87, and RMSE = 1060.58. The model explains approximately 58.6% of the variation in the validation target relative to the mean baseline.

The MAE indicates the typical absolute difference between the actual and predicted sales in the dataset's sales units. RMSE is higher than MAE, suggesting that some larger prediction errors influence the squared-error metric.

The supplied test file has no labels. Therefore, a true test accuracy or test R^2 cannot be calculated from the supplied files. The generated Test_Predictions.csv is a prediction artifact, not a scored evaluation.

The charts in the results folder show actual versus predicted values and the validation residual distribution.

10. Testing and Validation

Data checks: confirm that the target exists in the training file, inspect null counts and verify the test file has the same predictor columns.

Preprocessing checks: median and most-frequent imputers are fitted on training data only; one-hot encoding tolerates previously unseen categories.

Model checks: ensure the fitted regressor returns finite numerical predictions and compute R^2, MAE and RMSE on the hold-out partition.

Output checks: verify that the number of prediction rows equals the number of rows in Test.csv and that the output CSV can be reopened with pandas.

11. Limitations and Risk Considerations

The validation score depends on the particular random split and may differ from performance on future stores or time periods. A random split is not equivalent to a time-based or store-based external validation.

The supplied data may contain historical biases, missing context and changes in consumer behavior. Important factors such as promotions, local events, competitor pricing and seasonality may not be represented.

The current model is a baseline implementation. Hyperparameter tuning, cross-validation, feature engineering and error analysis by outlet/item segment may improve performance.

Predictions are estimates and should not be treated as guaranteed sales or used for high-impact business decisions without further testing.

12. Conclusion

The project demonstrates an end-to-end supervised regression workflow for Big Mart sales prediction. It loads the supplied data, handles missing values, encodes categorical attributes, trains a gradient boosting model and evaluates the result on held-out data.

The experiment achieved a validation R^2 of 0.5861. The model also generated sales estimates for all rows in the supplied test file. The deliverables include code, notebook, metrics, figures, documentation and presentation.

13. Future Scope

Compare the model with a mean baseline, Random Forest, XGBoost and CatBoost where installation is permitted.

Use cross-validation and systematic hyperparameter search to improve stability.

Add engineered features such as item age, outlet age, item visibility ratios and grouped item categories where justified by data.

Use time-aware validation if transaction dates become available. Add a lightweight dashboard for interactive sales estimation and monitoring.

APPENDIX A – HOW TO RUN

16. 1. Install Python 3.10 or later.
17. 2. Open a terminal in the project folder.
18. 3. Run: pip install -r requirements.txt
19. 4. Run the relevant script from src/ or open the notebook in Jupyter / Google Colab.
20. 5. Review the generated outputs in results/.
21. 6. Use the README for the complete repository workflow.

REFERENCES

22. Python Software Foundation. Python Documentation. https://docs.python.org/3/
23. scikit-learn Developers. scikit-learn User Guide. https://scikit-learn.org/stable/user_guide.html
24. Pandas Development Team. pandas Documentation. https://pandas.pydata.org/docs/
25. Matplotlib Development Team. Matplotlib Documentation. https://matplotlib.org/stable/
26. BigMart Sales dataset supplied with the internship project materials (Train.csv and Test.csv).

APPENDIX B – DATA DICTIONARY

This appendix documents the meaning and role of the main fields used in the project. Input features describe each observation. The target field is the value the supervised model learns to predict. Data types and target encoding must be preserved when loading the data. Review the README and notebook for the exact column names and label mapping.

APPENDIX C – ALGORITHM WALKTHROUGH

The selected estimator learns patterns from labelled training observations. During fitting, model parameters are adjusted to reduce the training objective. Hyperparameters such as architecture, iteration count, learning rate and regularization influence model complexity and generalization. The chosen configuration is a reproducible baseline rather than a claim of globally optimal performance.

APPENDIX D – PREPROCESSING DETAILS

Preprocessing converts raw data into a representation suitable for a learning algorithm. Missing values are handled using statistics learned from training data. Categorical fields, where present, are encoded into model-readable indicators. Numerical scaling is used in the neural-network project. All fitted transformations are applied consistently to validation and future records.

APPENDIX E – TEST CASES

Test 1: load the supplied or documented dataset and verify its dimensions. Test 2: inspect target labels and feature columns. Test 3: perform the train/validation split with the documented random seed. Test 4: fit preprocessing and the estimator using training data only. Test 5: generate predictions for held-out rows. Test 6: calculate the metrics and confirm that output files can be reopened.

APPENDIX F – RESULT INTERPRETATION

Evaluation metrics summarize performance on the held-out validation sample. They are not guarantees of future performance. Always interpret a metric with its definition, units, class mapping and validation design. Visualizations should be read alongside the numeric metrics, not as substitutes for them. The results folder contains the plots produced by the experiment.

APPENDIX G – GLOSSARY

Feature: an input variable. Target: the value to be predicted. Training set: observations used to fit the model. Validation set: held-out observations used to estimate generalization. Imputation: replacement of missing values. Overfitting: learning training-specific patterns that do not generalize. Reproducibility: ability to repeat an experiment using recorded data, code and settings.

APPENDIX H – REPRODUCIBILITY AND REPOSITORY

Keep data files, scripts, notebooks, reports, presentation and results in the documented folder structure. Install packages from requirements.txt. Run the Python script or execute the notebook from top to bottom. Keep the fixed random seed unchanged when reproducing the documented metrics. Do not upload virtual environments, cache files or private credentials to GitHub.

APPENDIX I – BUSINESS AND DATA LIMITATIONS

Sales estimates may be affected by changes in prices, promotions, consumer preferences, supply conditions and outlet characteristics. Historical correlations do not prove causation. Review errors by outlet and product category, validate against future data and retain human review before using estimates in business planning.